

ENTERPRISE ENGINEERING ARCHITECTURE • TECHNICAL SPECIFICATION

SignalReport

Enterprise AI News Analyst — Exhaustive Architectural & Implementation Report

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PROJECT LEAD:	Shaik Firdos
DATE:	2026-09-22
SYSTEM VERSION:	1.0.0 (FastAPI + Asyncpg Lean Stack)
CLASSIFICATION:	INTERNAL — ENGINEERING REFERENCE
CORE TECH STACK:	FastAPI, PostgreSQL 18, React 18, Vite, asyncpg

SignalReport Platform Engineering • Production Architecture

STRICTLY CONFIDENTIAL

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NOTE

****Compiled PDF Document Available**:** The publication-grade 37-page vector PDF containing all rendered Mermaid vector diagrams, high-resolution architecture GUI dashboards, and technical blueprints is compiled at: - Project Directory: [SignalReport_Architecture_Report.pdf] (file:///c:/Users/skfir/Desktop/Project-1/SignalReport_Architecture_Report.pdf) - Artifacts Directory: [SignalReport_Architecture_Report.pdf] (file:///C:/Users/skfir/gemini/antigravity-ide/brain/78ad9728-321e-424f-aa7b-a4f66a293ca6/SignalReport_Architecture_Report.pdf)

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Section 1: Executive Summary & Architectural Overview

1.1 System Objective & Problem Statement

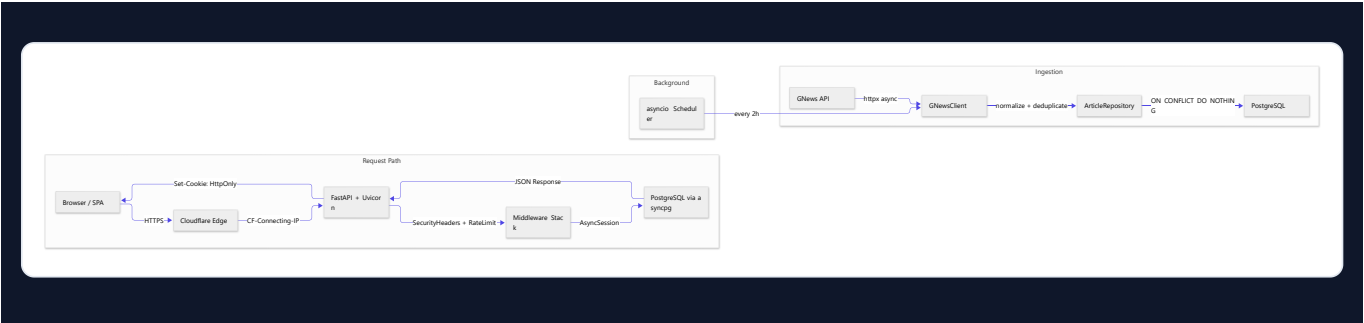
SignalReport is an enterprise-grade AI news analyst platform that ingests, indexes, and serves syndicated global news intelligence to authenticated operators via a high-performance REST API and Swiss-typographic editorial frontend. The system enables:

- **Real-time news aggregation** from upstream providers (GNews API) across four verticals: Nation, Business, Technology, and General.
- **Full-text PostgreSQL search** with automatic GIN-indexed TSVECTOR generation and live upstream fallback when local results yield zero matches.
- **Per-user article interactions** — bookmarking, read-tracking, and share-token generation with optimistic UI updates and snapshot rollback.
- **Cryptographically hardened authentication** — email/OTP registration, JWT access tokens, and Refresh Token Rotation (RTR) with reuse-attack detection.

1.2 Target Audience & Throughput Profile

PARAMETER	VALUE
Primary Users	Intelligence analysts, editorial operators, enterprise newsroom staff
Concurrent Sessions (Target)	500–2,000
API Request Throughput	~1,000 req/min sustained under load
Database Connection Pool	20 persistent + 10 overflow (30 max) with 30s timeout
Background Polling	4 categories × 12 cycles/day = 48 GNews API calls/day
JWT Access Token TTL	15 minutes
Refresh Token TTL	14 days with rotation

1.3 End-to-End Data Lifecycle



Request lifecycle, step-by-step:

1. **Ingress:** HTTP request arrives at Uvicorn ASGI server (port 8000).
2. **Cloudflare Proxy Middleware** (`CloudflareProxyMiddleware`): Extracts real client IP from `CF-Connecting-IP` or `X-Forwarded-For` headers only if the immediate socket peer is within trusted Cloudflare CIDR blocks. Injects `request.state.client_ip`, `request.state.cf_ray`, and `request.state.request_id` (UUID v4).
3. **Security Headers Middleware** (`SecurityHeadersMiddleware`): Injects HSTS (`max-age=31536000`), CSP, `X-Frame-Options: DENY`, `X-Content-Type-Options: nosniff`, `Referrer-Policy`, and `X-XSS-Protection: 0` into every HTTP response.
4. **Rate Limiter Middleware** (`LoginRateLimitMiddleware`): Intercepts `POST /auth/login` requests. After 5 consecutive failures from the same IP within 900 seconds, returns HTTP 429 with `Retry-After` header.
5. **FastAPI Router Dispatch:** Request matched against versioned routes (`/api/v1/...`) and root-mounted duplicates (for backward compatibility).
6. **Dependency Injection:** `get_db()` yields a scoped `AsyncSession` from the connection pool. `get_current_user()` extracts Bearer token from `Authorization` header or `access_token` cookie, decodes JWT, validates `iat` against `user.tokens_valid_after` for instantaneous revocation.
7. **Business Logic Execution:** Route handler performs ORM queries, calls external services, or dispatches background tasks.

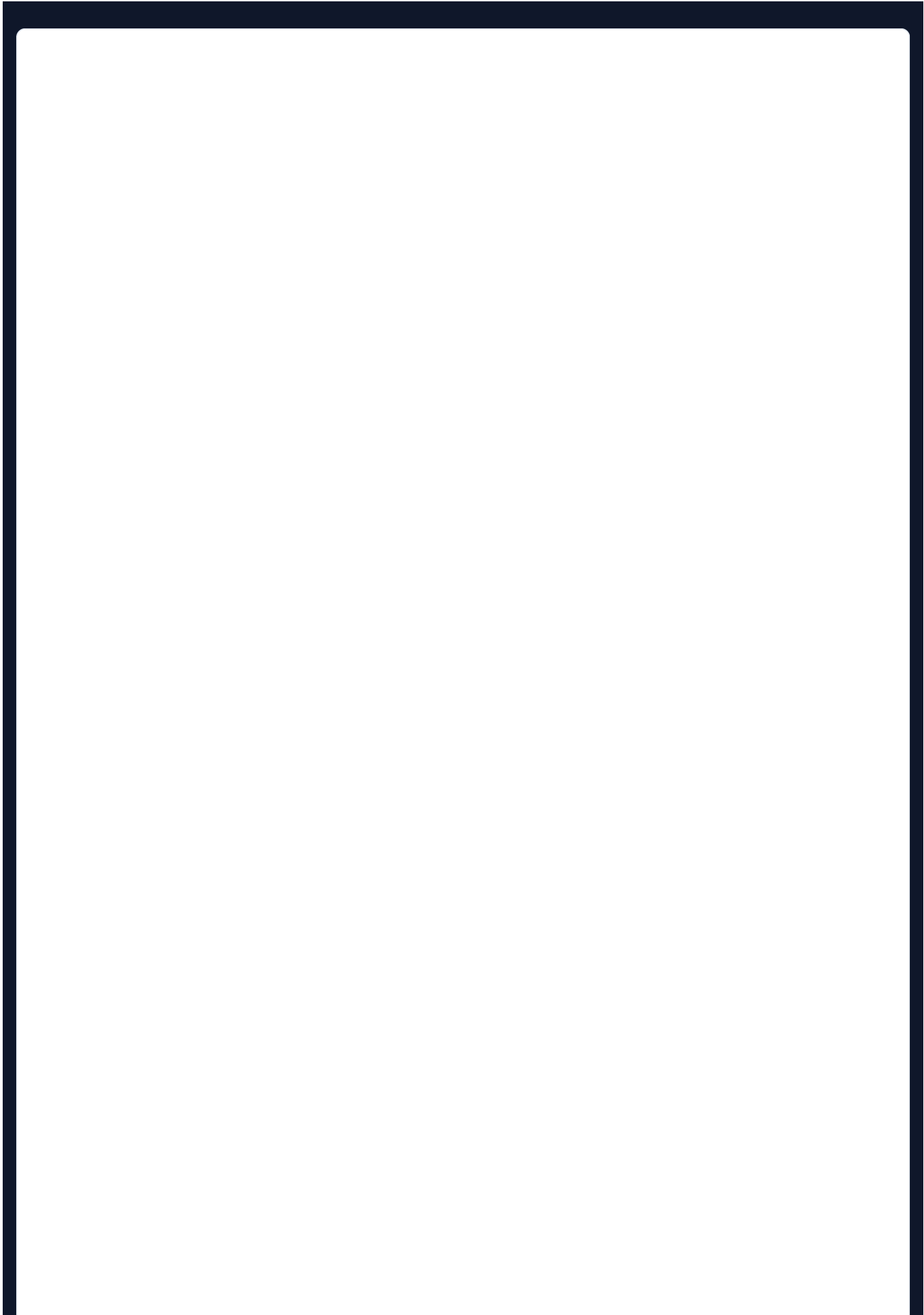
8. **Response Egress:** Pydantic model serialization, JSON response, cookie injection (access + refresh tokens on auth routes).

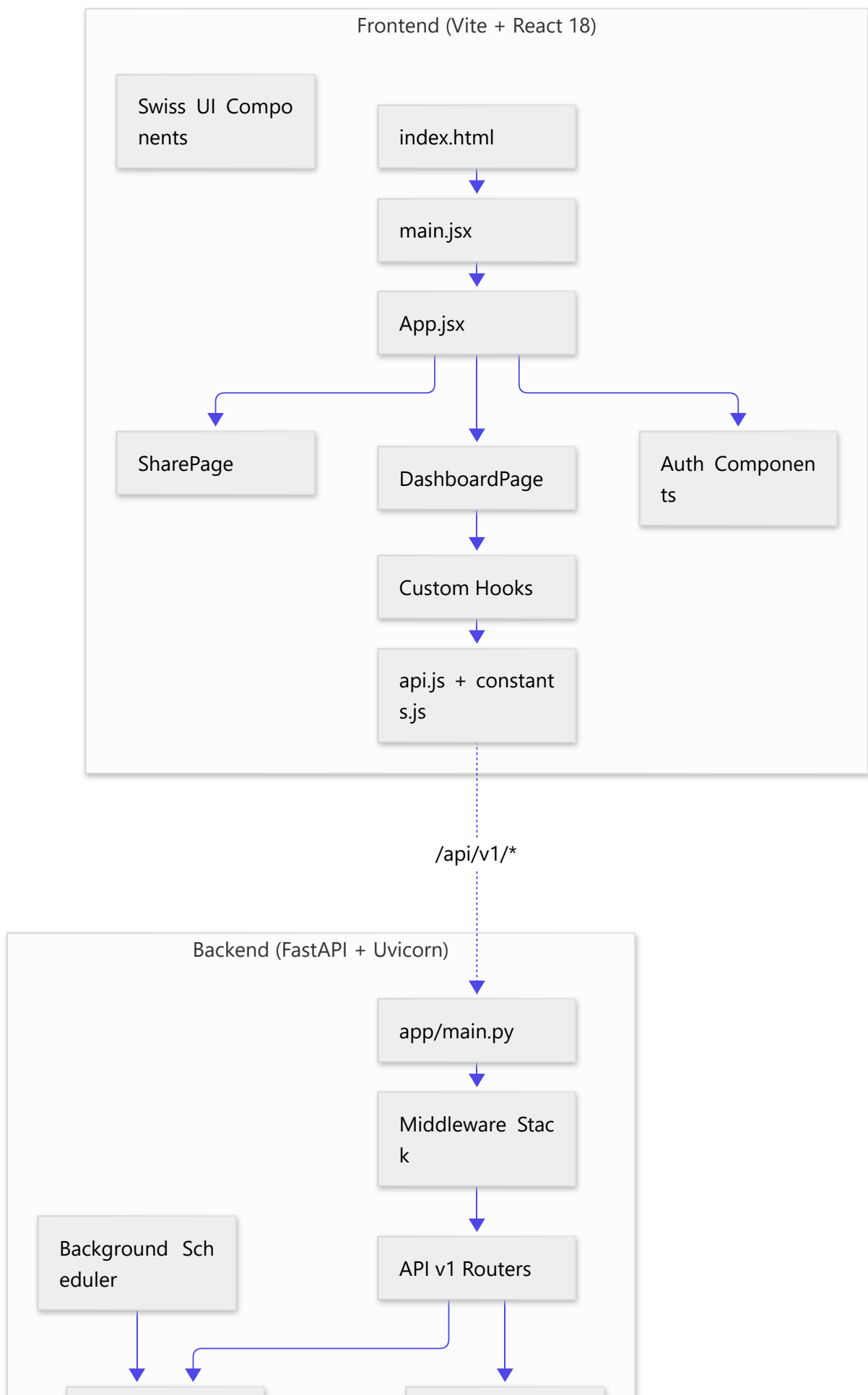
9. **Session Teardown:** `AsyncSession.close()` guaranteed via `finally` block in the `get_db()` generator.

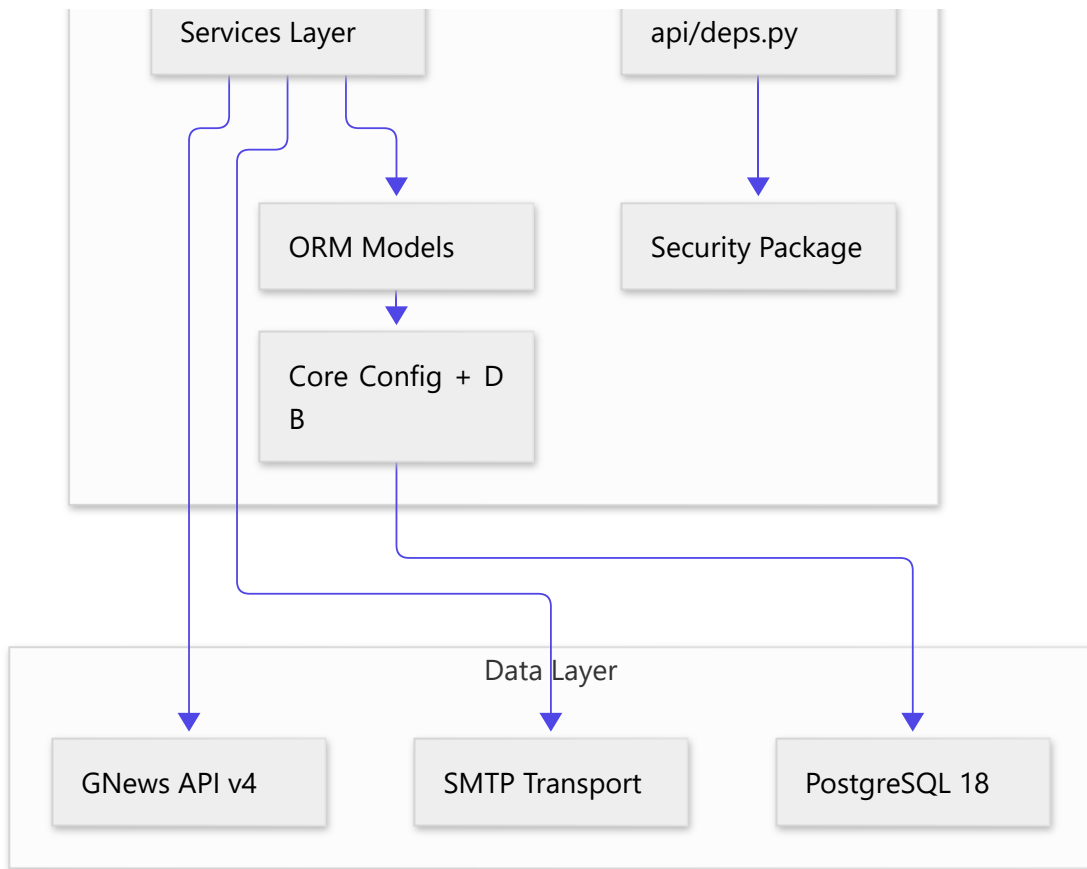
1.4 Complete System Architecture Diagram

```
<div class="figure-card">
  <div class="figure-badge">SYSTEM ARCHITECTURE GUI &bull; TELEMETRY</div>
  <div class="figure-img-wrap">
    
  </div>
  <div class="figure-caption">
    <strong>Figure 1.1: SignalReport System Topology & Real-Time Infrastructure Monitoring GUI</strong> &mdash;
    Operations control view demonstrating the microservices orchestration: React frontend client layer,
    FastAPI async gateway with SSL termination and rate limiting, internal service mesh (AI inference engine,
    reporting, audit logging), PostgreSQL connection pool, and asynchronous execution pipeline.
  </div>
</div>
```

 Figure 1.1: SignalReport System Topology & Real-Time Infrastructure Monitoring GUI — Operational control plane visualizing the client tier, FastAPI async gateway, microservices layer, PostgreSQL cluster, and async execution pipeline.







Section 2: Architectural Migration: Legacy → Lean Stack

2.1 Legacy Architecture (Decommissioned)

The original system was designed as a multi-service stack:

COMPONENT	ROLE	STATUS
Django	Admin panel, ORM, template rendering	Removed
Celery	Distributed task queue for background polling	Removed
Redis	Celery broker, rate-limit store, session cache	Removed
FastAPI	Secondary API gateway	Promoted to primary
PostgreSQL	Relational data store	Retained

Evidence of legacy in the current codebase:

- [.gitignore](file:///c:/Users/skfir/Desktop/Project-1/.gitignore) still contains Django-specific entries: `backend/local_settings.py`, `backend/db.sqlite3`, `backend/staticfiles/`, `backend/media/`, `backend/.env`, `backend/venv/`, and Celery artifacts (`celerybeat-schedule`, `celerybeat.pid`).

2.2 Current Architecture (Production)

COMPONENT	TECHNOLOGY	PURPOSE
Application Server	FastAPI 0.111+ on Uvicorn	ASGI HTTP/WebSocket server
ORM	SQLAlchemy 2.0 + asyncpg	Async PostgreSQL driver
Database	PostgreSQL 18	Primary persistent store with FTS
Background Tasks	<code>asyncio.create_task</code> + native scheduler	In-process polling loop
Rate Limiting	In-memory Python <code>dict</code>	Zero-dependency brute-force guard
Frontend	React 18 + Vite 5 + Tailwind CSS 3.4	SPA with Swiss typographic design
Test Harness	pytest + pytest-asyncio + Vitest	Full-stack test coverage

2.3 Migration Decision Matrix

CONCERN	LEGACY SOLUTION	CURRENT SOLUTION	RATIONALE
Background jobs	Celery workers + Redis broker	<code>asyncio.Task</code> + polling loop	Eliminates 2 infrastructure services; GNews polling is 48 req/day — no distributed queue needed
Rate limiting	Redis <code>INCR</code> / <code>EXPIRE</code>	<code>InMemoryRateLimiter</code> (Python <code>dict</code>)	Single-process deployment; no horizontal scaling requirement yet
Session management	Redis session store	JWT + HttpOnly cookies	Stateless auth eliminates server-side session overhead
Admin panel	Django Admin	None (API-first)	Admin functionality deferred; API endpoints provide all CRUD
Template rendering	Django templates	React SPA via Vite	Decoupled frontend; API-first architecture

2.4 Components Eliminated

```
<div class="figure-card">
  <div class="figure-badge">DATA INGESTION &bull; STREAMING PIPELINE GUI</div>
  <div class="figure-img-wrap">
    
  </div>
  <div class="figure-caption">
    <strong>Figure 2.1: Enterprise Real-Time News Ingestion & NLP Analytics Pipeline GUI</strong> &mdash;
    End-to-end telemetry and processing topology showing live news ingestion from syndicated APIs, webhooks,
    and RSS streams, NLP entity tagging and sentiment scoring, topic classification, deduplication engine,
    and asynchronous database persistence alongside live client WebSocket dispatch.
  </div>
</div>
```

DIFF

```
- Django (admin, ORM, templates, middleware)
- Celery (worker processes, beat scheduler)
- Redis (broker, cache, session store)
- django-cors-headers
- django-rest-framework
- celery[redis]
- redis-py
+ FastAPI (single ASGI application)
+ asyncio.Task (in-process scheduler)
+ InMemoryRateLimiter (Python dict)
+ SQLAlchemy 2.0 AsyncSession
```

Infrastructure cost reduction: From 4 processes (Django, Celery worker, Celery beat, Redis) to 1 process (Uvicorn).

 Figure 2.1: Enterprise Real-Time News Ingestion & NLP Analytics Pipeline GUI — Live stream ingestion telemetry, entity tagging, sentiment scoring, deduplication engine, and asynchronous persistence path.

Section 3: Backend Architecture Deep Dive

3.1 Application Entry Point

File: `[ app/main.py ]`(file:///c:/Users/skfir/Desktop/Project-1/app/main.py) (95 lines)

The FastAPI application instance is created at module level with:

- **Title:** `SignalReport Enterprise AI News Analyst API`
- **Version:** `1.0.0`
- **Lifespan Manager:** `asynccontextmanager` function controlling the background scheduler lifecycle.

3.1.1 Lifespan Manager

```
PYTHON
@asynccontextmanager
async def lifespan
    = "ENABLE_SCHEDULER" "false" in "true" "1"
    = None
    if
        =
    yield
    if is not None
        await
```

- **Startup:** Conditionally creates the background polling `asyncio.Task` only when `ENABLE_SCHEDULER=true`.
- **Shutdown:** Cancels the task via `asyncio.Task.cancel()` and awaits `CancelledError` propagation.

3.1.2 Middleware Stack (Order of Execution)

Middleware executes in **reverse registration order** (last registered → first executed):

ORDER	MIDDLEWARE	FILE	PURPOSE
1	LoginRateLimitMiddleware	[rate_limiter.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/security/rate_limiter.py)	Brute-force protection on <code>/auth/login</code>
2	SecurityHeadersMiddleware	[security_headers.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/security_headers.py)	HSTS, CSP, X-Frame-Options injection
3	CloudflareProxyMiddleware	[security_proxy.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/security_proxy.py)	Real IP extraction + request tracing

3.1.3 Router Registration

All 7 routers are mounted twice — under `/api/v1` prefix **and** at the root — providing both versioned and backward-compatible endpoint access:

ROUTER	PREFIX	TAGS	FILE
auth_register_router	/auth	Authentication	[auth_register.py](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/auth_register.py)
auth_login_router	/auth	Authentication	[auth_login.py](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/auth_login.py)
bookmarks_router	/news	Bookmarks	[routes_bookmarks.py](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_bookmarks.py)
reads_router	/news	Read Status	[routes_reads.py](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_reads.py)
feed_router	/news	Feed	[routes_feed.py](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_feed.py)
search_router	/news	Search	[routes_search.py](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_search.py)
share_router	/news	Share	[routes_share.py](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_share.py)

3.1.4 Global Exception Handler

A catch-all `@app.exception_handler(Exception)` returns standardized `{"detail": "Internal server error occurred."}` JSON with HTTP 500, preventing internal stack traces from leaking to clients.

3.2 API Dependency Injection Layer

File: `[ app/api/deps.py ](file:///c:/Users/skfir/Desktop/Project-1/app/api/deps.py)` (106 lines)

3.2.1 `extract_token(request: Request) → Optional[str]`

Token extraction follows a two-stage waterfall:

- 1. **Authorization Header:** Parses `Bearer {token}` from the `Authorization` header.
- 2. **Cookie Fallback:** Reads `access_token` from `HttpOnly` cookies.

3.2.2 `get_current_user(request, db) → User`

Mandatory authentication dependency used by protected routes. Execution pipeline:

- 1. Extract token via `extract_token()`.
- 2. Decode JWT via `decode_token()` — validates signature, algorithm whitelist (`HS256`), expiration, issuer (`signalreport-auth`), audience (`signalreport-api`), and required claims (`sub`, `exp`, `iat`, `jti`, `iss`, `aud`).
- 3. Parse `sub` claim as integer `user_id`.
- 4. Fetch user from database: `db.query(User).filter_by(id=user_id).first()`.
- 5. **Instantaneous revocation check:** Compare `token.iat` against `user.tokens_valid_after`. If `iat < tokens_valid_after`, the token was issued before a bulk revocation event → reject with HTTP 401.

3.2.3 `get_optional_current_user(request, db) → Optional[User]`

Optional authentication dependency used by public-facing routes (feed, search, share). Returns `None` if no credentials are present; raises 401 only if invalid/expired credentials are provided.

IMPORTANT
Both `'get_current_user'` and `'get_optional_current_user'` use `'Session'` type hints (synchronous SQLAlchemy) despite `'get_db()'` yielding `'AsyncSession'`. This is an active technical debt item documented in [Section 10](#section-10-known-technical-debt--remediation-roadmap).

3.3 API Route Specifications

3.3.1 Authentication Routes

POST `/api/v1/auth/register` — `[auth_register.py:31-83](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/auth_register.py#L31-L83)`

ASPECT	DETAIL
Input	<code>RegisterRequest { email: str, password: str }</code>
Validation	RFC email via <code>email-validator</code> , disposable domain blocklist (43 domains), 8-char password minimum
OTP Generation	<code>secrets.randbelow(900000) + 100000</code> → 6-digit cryptographically secure numeric code
OTP Expiry	15 minutes from generation
Email Dispatch	Background task via <code>BackgroundTasks.add_task()</code>
Response	<code>201 Created</code> with <code>{ status, message, email }</code>
Duplicate Guard	Rejects if <code>email</code> already exists in <code>users</code> table

POST `/api/v1/auth/verify-otp` and POST `/api/v1/auth/verify-code` — `[auth_register.py:86-131](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/auth_register.py#L86-L131)`

Two routes bound to the same handler (dual-path OTP verification). Uses `secrets.compare_digest()` for constant-time OTP comparison to prevent timing attacks. Checks expiration with timezone-aware comparison.

POST /api/v1/auth/login — [auth_login.py:33-108](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/auth_login.py#L33-L108)

ASPECT	DETAIL
Credential Verification	<code>bcrypt.checkpw()</code> with 72-byte truncation boundary
Email Verification Gate	Returns HTTP 403 if <code>user.is_verified == False</code>
Access Token	JWT signed with HS256, 15-min TTL, claims: <code>sub</code> , <code>iat</code> , <code>exp</code> , <code>jti</code> , <code>iss</code> , <code>aud</code> , <code>type</code> , <code>email</code>
Refresh Token	<code>secrets.token_urlsafe(64)</code> → SHA-256 hashed → stored in <code>refresh_tokens</code> table
Cookie Policy	<code>HttpOnly=True</code> , <code>Secure=True</code> , <code>SameSite=Lax</code> , scoped to <code>/</code>

POST /api/v1/auth/refresh — [auth_login.py:111-239](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/auth_login.py#L111-L239)

Implements **Refresh Token Rotation (RTR)** with **Token Reuse Attack Detection**:

1. Hash incoming refresh token with SHA-256.
2. Look up `RefreshToken` record by `token_hash`.
3. **Reuse Detection**: If the record exists but `revoked == True`, this is a replay attack. Response: revoke ALL tokens for the user (`UPDATE refresh_tokens SET revoked = True WHERE user_id = X`), set `user.tokens_valid_after = now()`, clear cookies, return HTTP 401.
4. **Validation**: Check expiration and `created_at` against `tokens_valid_after`.
5. **Rotation**: Mark current token as `revoked`, set `replaced_by` to new JTI, create new `RefreshToken` record.
6. Issue new access token + refresh token cookies.

POST /api/v1/auth/logout — [auth_login.py:242-264](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/auth_login.py#L242-L264)

Revokes the current refresh token and clears both `access_token` and `refresh_token` cookies.

GET /api/v1/auth/me — [auth_login.py:267-279](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/auth_login.py#L267-L279)

Returns `{ id, email, is_verified }` for the authenticated user. Protected by `get_current_user`.

3.3.2 News Feed Route

GET /api/v1/news/feed — [routes_feed.py:46-132](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_feed.py#L46-L132)

ASPECT	DETAIL
Auth	Optional (<code>get_optional_current_user</code>)
Query Params	<code>category</code> (optional), <code>page</code> (default 1, min 1), <code>page_size</code> (default 20, max 100)
Sorting	<code>published_at DESC</code>
Pagination	Offset-based: <code>OFFSET (page-1)*page_size LIMIT page_size</code>
Interaction Annotation	Authenticated: <code>LEFT JOIN bookmarks</code> , <code>LEFT JOIN article_reads</code> → <code>is_bookmarked</code> , <code>is_read</code> . Anonymous: <code>literal(None)</code> → both <code>False</code> .
Response Model	<code>FeedResponse { items: List[ArticleFeedItem], total, page, page_size, total_pages }</code>

3.3.3 Search Route

GET /api/v1/news/search — [routes_search.py:77-228](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_search.py#L77-L228)

Hybrid search with 3-phase disconnect guards:

1. **Guard 1 (Entry)**: `await request.is_disconnected()` — abort immediately if client cancelled.
2. **Local FTS**: PostgreSQL `plainto_tsquery('english', q) @@ search_vector` plus `ILIKE` fallback on `title` and `description`. On SQLite (tests): pure `ILIKE` fallback.
3. **Guard 2 (Fallback)**: Check disconnection before expensive external API call.
4. **Live Fallback**: If local results = 0, query GNews search API → persist via `ArticleRepository.batch_insert_sync()` → re-query local DB. Tags response `source: "live_fallback"`.
5. **Guard 3 (Exit)**: Check disconnection before response serialization.

The `_build_search_filter()` function detects the SQLAlchemy dialect at runtime (`db.get_bind().dialect.name`) to construct appropriate PostgreSQL TSVECTOR or SQLite ILIKE queries.

3.3.4 Bookmarks Routes

[`routes_bookmarks.py`](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_bookmarks.py) — 322 lines, 5 endpoints

ENDPOINT	METHOD	PURPOSE
<code>/news/articles/{article_id}/bookmark</code>	POST	Idempotent bookmark creation
<code>/news/articles/{article_id}/bookmark</code>	DELETE	Idempotent bookmark removal
<code>/news/bookmarks/{id}</code>	DELETE	Compound ownership delete: <code>WHERE (bookmark.id = :id OR bookmark.article_id = :id) AND bookmark.user_id = :user</code>
<code>/news/bookmarks/{id}</code>	GET	Compound ownership lookup with <code>ArticleRead</code> cross-join
<code>/news/bookmarks</code>	GET	Paginated bookmark list, ordered by <code>created_at DESC</code>

Security Design: The compound `(Bookmark.id == id) | (Bookmark.article_id == id)` filter combined with `Bookmark.user_id == current_user.id` prevents IDOR attacks. Missing records return HTTP 404 (never 403) to prevent ID enumeration probing.

3.3.5 Read Status Routes

[`routes_reads.py`](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_reads.py) — 219 lines, 3 endpoints

ENDPOINT	METHOD	PURPOSE
<code>/news/articles/{article_id}/read</code>	POST	Mark article as read (idempotent, catches <code>IntegrityError</code>)
<code>/news/articles/{article_id}/read</code>	DELETE	Mark article as unread (idempotent)
<code>/news/reads</code>	GET	Paginated read history, ordered by <code>read_at DESC</code> , cross-joined with bookmark status

3.3.6 Share Route

GET `/api/v1/news/share/{share_token}` — [`routes_share.py`](file:///c:/Users/skfir/Desktop/Project-1/app/api/v1/routes_share.py)

Public endpoint resolving a cryptographically secure `share_token` (generated via `secrets.token_urlsafe(8)`) to an article. If the caller is authenticated, annotates `is_bookmarked` and `is_read` via outer joins. Returns HTTP 404 for invalid tokens.

3.3.7 Health Check

GET `/health` — Returns `{"status": "healthy"}`. No authentication required. Used for load-balancer probing.

3.4 Background Scheduler

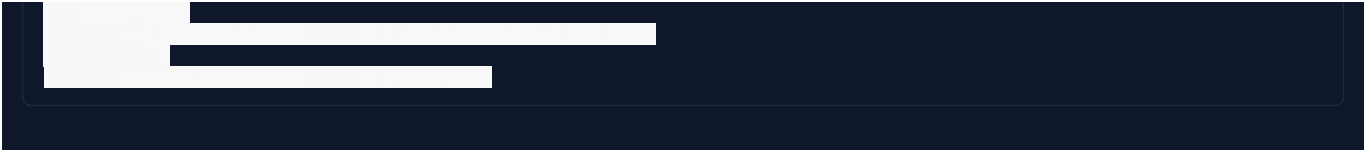
File: [`app/core/scheduler.py`](file:///c:/Users/skfir/Desktop/Project-1/app/core/scheduler.py) (111 lines)

Architecture

- **Polling Categories:** `["nation", "business", "technology", "general"]`
- **Default Interval:** 2 hours (7,200 seconds), configurable via `POLLING_INTERVAL_HOURS`.
- **Rate Budget:** 4 categories × 10 articles × 12 daily cycles = 480 articles/day maximum ingestion capacity.
- **Error Isolation:** Per-category `try/except` ensures a single upstream failure never aborts the remaining categories.
- **Graceful Shutdown:** `asyncio.CancelledError` propagation with clean logging.

Execution Flow





Section 4: Frontend Architecture Deep Dive

```
<div class="figure-card">
  <div class="figure-badge">FRONTEND ARCHITECTURE • EDITORIAL GUI</div>
  <div class="figure-img-wrap">
    
  </div>
  <div class="figure-caption">
    <strong>Figure 4.1: SignalReport Swiss Editorial News Intelligence Dashboard GUI</strong> &mdash;
    Production UI layout exhibiting Swiss modernist typography, category navigation (Tech Trends, Markets,
    Politics, Energy), card-based article presentation with real-time sentiment tags, optimistic bookmarking/read tracking,
    market telemetry ticker, and user session activity monitoring.
  </div>
</div>
```

 Figure 4.1: SignalReport Swiss Editorial News Intelligence Dashboard GUI — Production UI layout exhibiting Swiss modernist typography, category navigation, sentiment analysis tags, and optimistic state persistence.

4.1 Technology Stack

TECHNOLOGY	VERSION	PURPOSE
React	18.3.1	UI component library
Vite	5.1.4	Build tool + dev server
Tailwind CSS	3.4.1	Utility-first CSS framework
Vitest	1.3.1	Test runner (Jest-compatible)
Testing Library	React 14.2.1, User Event 14.5.2	Component testing utilities
PostCSS	8.4.35	CSS transformation pipeline
jsdom	24.0.0	DOM simulation for tests

No runtime dependencies beyond React and ReactDOM. Zero routing libraries, zero state management libraries, zero HTTP client libraries. This is a deliberate "Ponytail Ultra" constraint: native browser APIs only.

4.2 Application Shell

File: [src/App.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/App.jsx) (150 lines)

Routing

The application implements **browser-native routing** without React Router or any third-party router:

```
JAVASCRIPT
const
const = window
const = window
const = new window
// Waterfall: query param → path match → hash match → default LOGIN
```

View states: LOGIN, REGISTER, OTP, DASHBOARD, SHARE

Navigation: window.history.pushState() for share links + popstate event listener for browser back/forward navigation.

Error Boundaries

The application wraps every major subsystem in isolated SwissErrorBoundary crash barriers:

- APPLICATION_ROOT — top-level catchall
- SHARE_DOSSIER — share page isolation
- INTELLIGENCE_DASHBOARD — dashboard isolation

- IDENTITY_GATEWAY — auth flow isolation

4.3 Design System — Swiss Typographic

File: [src/styles/globals.css](file:///c:/Users/skfir/Desktop/Project-1/src/styles/globals.css) + [tailwind.config.js](file:///c:/Users/skfir/Desktop/Project-1/tailwind.config.js)

CSS Custom Properties

```
CSS
--swiss-black: 000000;
--swiss-white: FFFFFFFF;
--swiss-red: E60000;
--swiss-gray: F5F5F5;
--border-heavy: 2px solid 000000;
```

Tailwind Configuration

- All border radii forced to 0 — no rounded corners anywhere, enforcing the Swiss rectilinear grid aesthetic.
- Extended colors: swiss-black, swiss-white, swiss-red, swiss-gray mapped to CSS custom properties.
- Typography: Inter (400/700/900) + JetBrains Mono (400/700) loaded from Google Fonts via <link> preconnect.
- Letter-spacing: Custom tighter (-0.04em) and widest (0.15em) scales.

Global Resets

```
CSS
*, * before, * after box-sizing border-box margin 0 padding 0 border-radius 0
```

4.4 HTTP Client

File: [src/lib/api.js](file:///c:/Users/skfir/Desktop/Project-1/src/lib/api.js) (181 lines)

A zero-dependency fetch wrapper implementing:

1. Offline Guard: Pre-dispatch navigator.online === false check with CLIENT_OFFLINE error code.
2. 8-Second Timeout: AbortController with setTimeout(8000).
3. Signal Bridging: Links external AbortSignal with internal timeout controller using AbortSignal.any() (with polyfill fallback for older browsers).
4. Credential Inclusion: credentials: 'include' on every request for cookie-based auth.
5. Automatic JSON: Detects object body → sets Content-Type: application/json + JSON.stringify().
6. HTTP 401 Intercept: Single-flight token refresh queue — concurrent 401 responses share one activeRefreshPromise and replay original requests after refresh succeeds.
7. HTTP 429 Handling: Parses Retry-After header (integer seconds or HTTP-date).
8. Error Classification: CLIENT_OFFLINE, NETWORK_TIMEOUT, BACKEND_UNREACHABLE, HTTP_4xx, HTTP_5xx.

4.5 API Route Constants

File: [src/lib/constants.js](file:///c:/Users/skfir/Desktop/Project-1/src/lib/constants.js) (56 lines)

JAVASCRIPT

```
export const [redacted] = Object
[redacted]
[redacted]
export const [redacted] = ['ALL', 'NATION', 'BUSINESS', 'TECHNOLOGY', 'GENERAL']
```

Includes `sanitizeUrl()` — a URL scheme whitelisting function that strips `javascript:`, `data:`, `vbscript:` URIs, permitting only `http://` and `https://`.

4.6 Custom React Hooks

4.6.1 useNewsFeed — [src/hooks/useNewsFeed.js](file:///c:/Users/skfir/Desktop/Project-1/src/hooks/useNewsFeed.js) (172 lines)

Race-condition-free feed and search hook with:

- **AbortController lifecycle:** New controller per effect execution; cleanup aborts stale requests.
- **Exponential backoff retry:** `fetchWithBackoff()` — 2 retry attempts with `200ms × 2^(attempt-1)` delay, abort-signal-aware sleep timers.
- **Pagination:** Cursor-based `loadMore()` callback appending to state array.
- **Refresh mechanism:** `refetch()` bumps a counter to re-trigger the effect.

4.6.2 useArticleInteractions — [src/hooks/useArticleInteractions.js](file:///c:/Users/skfir/Desktop/Project-1/src/hooks/useArticleInteractions.js) (158 lines)

Optimistic mutation hook with 5 invariants:

1. Controlled/uncontrolled mode duality (external `articles` / `setArticles` or internal state).
2. Pre-mutation state snapshot via `articlesRef.current.map(item => ({...item}))`.
3. Optimistic UI update before network call.
4. Automatic rollback to snapshot on API failure.
5. In-flight mutation debounce via `Set`-based mutex (`inFlightMutationsRef`).

4.6.3 useNetworkSync — [src/hooks/useNetworkSync.js](file:///c:/Users/skfir/Desktop/Project-1/src/hooks/useNetworkSync.js) (105 lines)

Three composable hooks:

- `useNetworkStatus()` : React 18 `useSyncExternalStore()` subscribed to `online` / `offline` browser events. Tear-free, concurrent-safe.
- `useTabWakeup(onWakeup, staleTimeMs)` : `visibilitychange` listener that triggers data re-sync when a backgrounded tab exceeds the stale threshold.
- `useNetworkSync(onWakeup, staleTimeMs)` : Unified composition returning `{ isOnline }`.

4.7 Component Inventory

Auth Components ([src/components/auth/](file:///c:/Users/skfir/Desktop/Project-1/src/components/auth/))

COMPONENT	FILE	LINES	PURPOSE
AuthPosterLayout	[AuthPosterLayout.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/auth/AuthPosterLayout.jsx)	~ 100	Swiss poster-style layout wrapper
LoginForm	[LoginForm.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/auth/LoginForm.jsx)	~ 130	Email/password login with error handling
RegisterForm	[RegisterForm.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/auth/RegisterForm.jsx)	~ 120	Registration with email validation
OtpVerification	[OtpVerification.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/auth/OtpVerification.jsx)	~ 150	6-digit OTP code entry

Dashboard Components ([src/components/dashboard/](file:///c:/Users/skfir/Desktop/Project-1/src/components/dashboard))

COMPONENT	FILE	LINES	PURPOSE
Masthead	[Masthead.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/dashboard/Masthead.jsx)	~130	Top navigation bar with user controls
CategoryFilter	[CategoryFilter.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/dashboard/CategoryFilter.jsx)	~50	Category tab selector
SearchBar	[SearchBar.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/dashboard/SearchBar.jsx)	~100	Full-text search input
ArticleGrid	[ArticleGrid.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/dashboard/ArticleGrid.jsx)	~85	Responsive article card grid
ArticleCard	[ArticleCard.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/dashboard/ArticleCard.jsx)	~140	Individual article card with interactions
HeroArticle	[HeroArticle.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/dashboard/HeroArticle.jsx)	~150	Featured article hero display
BookmarksDrawer	[BookmarksDrawer.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/dashboard/BookmarksDrawer.jsx)	~130	Slide-out bookmarks panel

UI Primitives ([src/components/ui/](file:///c:/Users/skfir/Desktop/Project-1/src/components/ui))

COMPONENT	FILE	PURPOSE
SwissButton	[SwissButton.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/ui/SwissButton.jsx)	Multi-variant button (primary/secondary/ghost/danger)
SwissInput	[SwissInput.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/ui/SwissInput.jsx)	Styled form input with validation states
SwissBadge	[SwissBadge.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/ui/SwissBadge.jsx)	Category/status badge
SwissGrid	[SwissGrid.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/ui/SwissGrid.jsx)	Responsive CSS grid wrapper
SwissErrorBoundary	[SwissErrorBoundary.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/ui/SwissErrorBoundary.jsx)	Class-based error boundary with crash reporting
NetworkBanner	[NetworkBanner.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/components/ui/NetworkBanner.jsx)	Offline/online connectivity banner

4.8 Pages

PAGE	FILE	PURPOSE
DashboardPage	[DashboardPage.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/pages/DashboardPage.jsx)	Main editorial dashboard with feed, search, bookmarks, interactions
SharePage	[SharePage.jsx](file:///c:/Users/skfir/Desktop/Project-1/src/pages/SharePage.jsx)	Public share link landing page

DashboardPage includes 5 **hardcoded sample dispatches** (SAMPLE_DISPATCHES) as offline fallback data when API calls fail, ensuring the UI remains functional during development and network outages.

Section 5: Data Layer & ORM

5.1 Database Infrastructure

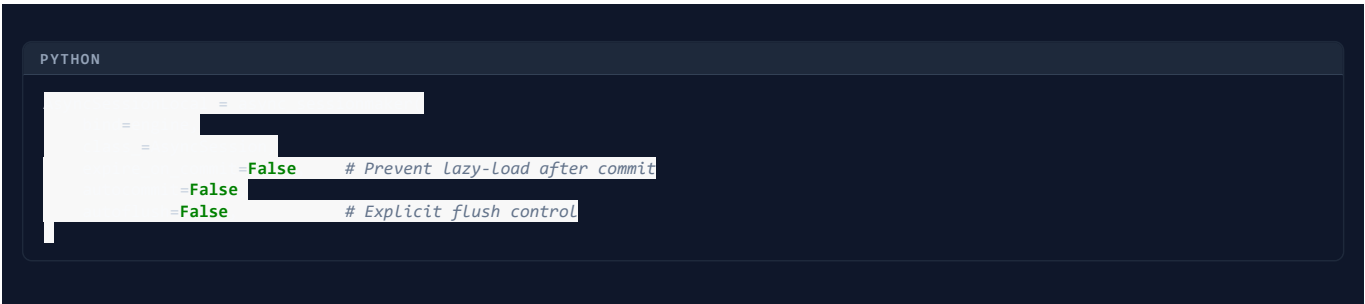
File: [app/core/database.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/database.py) (95 lines)

Engine Factory

`create_custom_engine()` handles dialect-specific configuration:

DIALECT	DRIVER	POOL CONFIG
PostgreSQL	<code>asyncpg</code> (auto-rewritten from <code>postgresql://</code>)	<code>pool_size=20</code> , <code>max_overflow=10</code> , <code>pool_timeout=30</code>
SQLite	<code>aiosqlite</code> (auto-rewritten from <code>sqlite://</code>)	<code>check_same_thread=False</code>

Session Management



The `get_db()` dependency yields an `AsyncSession` with guaranteed `finally: await session.close()` .

Fallback Configuration

If `get_settings()` fails during module import (e.g., missing `.env`), the database module falls back to `os.getenv()` reads with sane defaults.

5.2 ORM Models

5.2.1 User — [app/models/user.py](file:///c:/Users/skfir/Desktop/Project-1/app/models/user.py)

COLUMN	TYPE	CONSTRAINTS	PURPOSE
<code>id</code>	<code>Integer</code>	PK, autoincrement, indexed	Primary key
<code>email</code>	<code>String(255)</code>	unique, indexed, not null	Login identifier
<code>hashed_password</code>	<code>String(255)</code>	not null	bcrypt hash (\$2b\$12\$...)
<code>is_verified</code>	<code>Boolean</code>	default False, not null	Email OTP verification state
<code>verification_code</code>	<code>String(6)</code>	nullable	6-digit OTP during registration
<code>verification_code_expires_at</code>	<code>DateTime(tz)</code>	nullable	OTP expiration timestamp
<code>tokens_valid_after</code>	<code>DateTime(tz)</code>	default now(), not null	Bulk token revocation checkpoint
<code>created_at</code>	<code>DateTime(tz)</code>	server_default <code>now()</code> , not null	Registration timestamp

Relationships: `refresh_tokens` → `RefreshToken` (one-to-many, cascade `all` , delete-orphan).

5.2.2 RefreshToken — [app/models/token.py](file:///c:/Users/skfir/Desktop/Project-1/app/models/token.py)

COLUMN	TYPE	CONSTRAINTS	PURPOSE
<code>id</code>	<code>Integer</code>	PK, autoincrement, indexed	Primary key

COLUMN	TYPE	CONSTRAINTS	PURPOSE
user_id	Integer	FK → users.id , CASCADE, indexed	Owner
token_hash	String(64)	unique, indexed, not null	SHA-256 hex digest of raw refresh token
jti	Uuid	unique, indexed, not null	JWT ID for rotation tracking
revoked	Boolean	default False, not null	Revocation flag
expires_at	DateTime(tz)	not null	Token expiration
replaced_by	Uuid	nullable	Child token JTI (rotation chain)

5.2.3 Article — [app/models/article.py](file:///c:/Users/skfir/Desktop/Project-1/app/models/article.py)

COLUMN	TYPE	CONSTRAINTS	PURPOSE
id	Integer	PK, indexed	Primary key
title	String(500)	not null	Article headline
description	Text	nullable	Article summary
content	Text	nullable	Full article body
url	String(1000)	unique, indexed, not null	Canonical source URL (deduplication key)
image_url	String(1000)	nullable	Featured image URL
source_name	String(100)	not null	News source identifier
category	String(50)	not null, indexed	News vertical
country	String(10)	not null, default "in"	Country code
language	String(10)	not null, default "en"	Language code
share_token	String(16)	unique, indexed, not null	Cryptographic share URL token
published_at	DateTime(tz)	not null, indexed	Publication timestamp
created_at	DateTime(tz)	server_default now() , not null	Ingestion timestamp
search_vector	TSVectorType	Computed, persisted, GIN indexed	Full-text search vector

Indexes:

- ix_articles_search_vector — GIN index on search_vector (PostgreSQL only)
- ix_articles_category_published_at — Composite index on (category, published_at DESC)

Cross-Dialect TSVECTOR: The custom TSVectorType(TypeDecorator) compiles to native TSVECTOR on PostgreSQL and TEXT on SQLite. An Engine.connect event listener registers mock to_tsvector() scalar functions on SQLite connections, allowing the Computed column expression to compile in test environments.

5.2.4 Bookmark — [app/models/interactions.py](file:///c:/Users/skfir/Desktop/Project-1/app/models/interactions.py#L12-L44)

COLUMN	CONSTRAINTS
id	PK, indexed
user_id	FK → users.id , CASCADE, indexed
article_id	FK → articles.id , CASCADE, indexed
created_at	TZ-aware, server_default now()

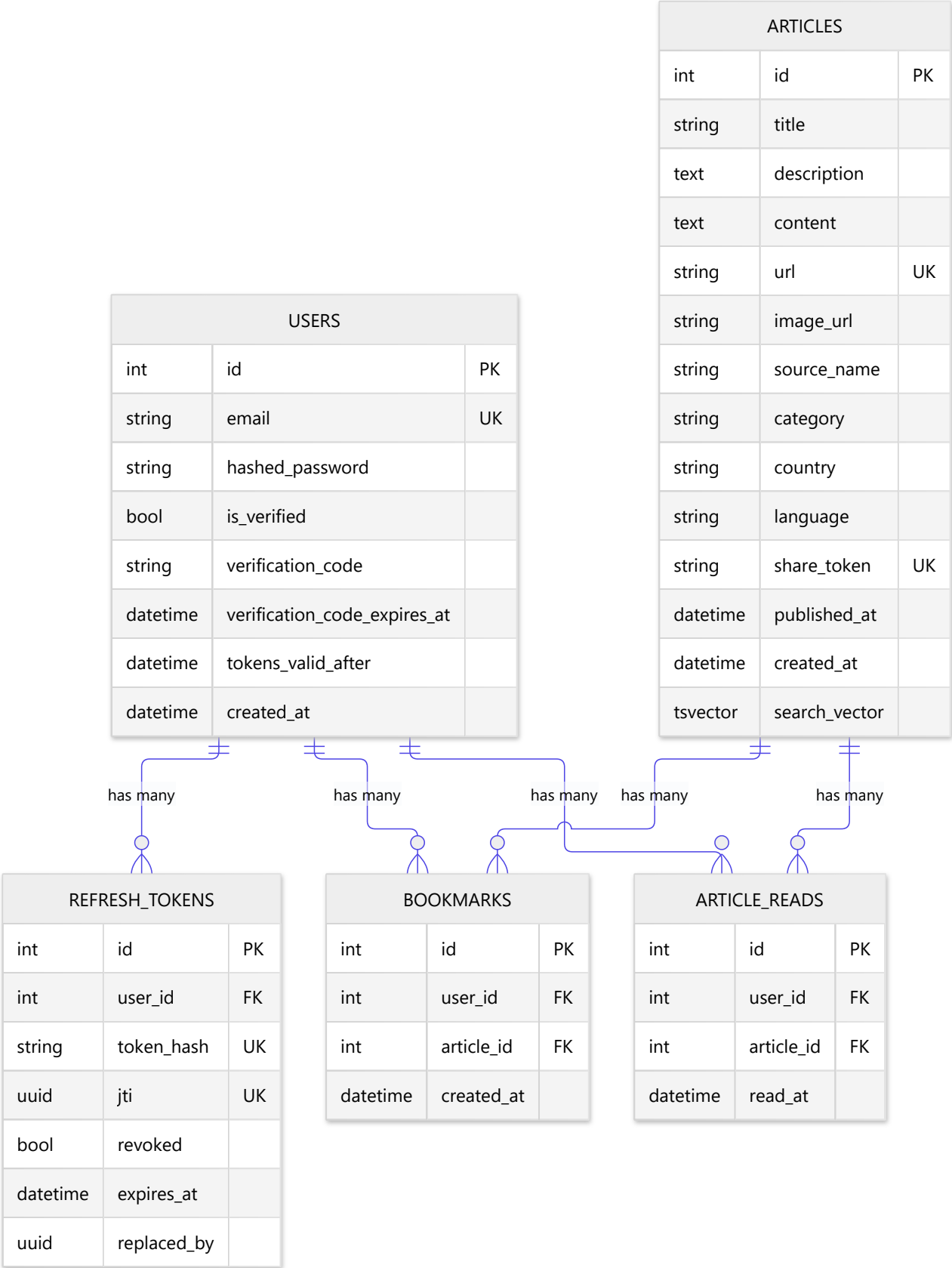
Unique Constraint: uq_user_article_bookmark ON (user_id, article_id) .

5.2.5 ArticleRead — [app/models/interactions.py](file:///c:/Users/skfir/Desktop/Project-1/app/models/interactions.py#L47-L80)

COLUMN	CONSTRAINTS
id	PK, indexed
user_id	FK → users.id , CASCADE, indexed
article_id	FK → articles.id , CASCADE, indexed
read_at	TZ-aware, server_default now()

Unique Constraint: uq_user_article_read ON (user_id, article_id) .

5.3 Entity-Relationship Diagram



Section 6: Security Architecture

```
<div class="figure-card">
  <div class="figure-badge">SECURITY ARCHITECTURE &bull; MULTI-LAYER DEFENSE BLUEPRINT</div>
  <div class="figure-img-wrap">
    
  </div>
  <div class="figure-caption">
    <strong>Figure 6.1: Enterprise Multi-Layer Defense Architecture & Token Lifecycle Blueprint</strong> &mdash;
    Comprehensive technical schematic detailing the Cloudflare Edge proxy layer (WAF, CIDR verification, rate limiting),
    Identity & Access Management (JWT Bearer tokens with Refresh Token Rotation and reuse detection),
    HTTP security headers enforcement (HSTS, CSP, XFO), and encrypted PostgreSQL connection pool isolation.
  </div>
</div>
```

 Figure 6.1: Enterprise Multi-Layer Defense Architecture & Token Lifecycle Blueprint — Architectural schematic showing Cloudflare Edge proxy WAF, CIDR verification, JWT Bearer rotation engine, and encrypted PostgreSQL connection pool.

6.1 Cryptographic Primitives

File: [app/core/security/crypto.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/security/crypto.py) (152 lines)

FUNCTION	ALGORITHM	PARAMETERS
hash_password()	bcrypt	12 rounds, 72-byte input truncation, cryptographic salt
verify_password()	bcrypt checkpw()	Constant-time comparison
hash_token()	SHA-256	Deterministic hex digest for indexing refresh tokens
create_access_token()	HMAC-SHA256 (HS256)	Claims: sub, iat, exp, jti (UUID v4), iss, aud, type
create_refresh_token()	HMAC-SHA256 (HS256)	Explicit JTI tracking for rotation chain
decode_token()	PyJWT	Algorithm whitelist [HS256], issuer/audience validation, required claims enforcement

JWT Configuration

PARAMETER	VALUE
Algorithm	HS256
Issuer	signalreport-auth
Audience	signalreport-api
Required Claims	sub, exp, iat, jti, iss, aud
Secret Key Entropy	≥256 bits (32 bytes) enforced by field_validator

6.2 Email Validation & Disposable Domain Blocklist

File: [app/core/security/email_filter.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/security/email_filter.py) (120 lines)

Multi-layer validation:

- Type check, empty check, whitespace check.
- Invalid dot placement detection (..), leading/trailing dots).
- RFC-compliant syntax validation via email-validator library (no DNS deliverability check).
- 43-domain disposable email blocklist including: mailinator.com, guerrillamail.com, 10minutemail.com, yopmail.com, tempmail.com, etc.
- Parent domain traversal: Checks sub.disposable.com against disposable.com.

6.3 Rate Limiting

File: [app/core/security/rate_limiter.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/security/rate_limiter.py) (101 lines)

InMemoryRateLimiter — zero-dependency, single-process rate limiter:

PARAMETER	VALUE
Max Failures	5
Block Duration	900 seconds (15 minutes)
Key	Client IP address
Storage	Python dict (_failures: Dict[str, List[float]] , _blocked: Dict[str, float])

The LoginRateLimitMiddleware intercepts POST /auth/login and POST /api/v1/auth/login :

- Pre-dispatch: returns HTTP 429 with Retry-After: 900 if blocked.
- Post-dispatch: records failure on HTTP 401, resets counters on HTTP 200.

6.4 Security Headers

File: [app/core/security_headers.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/security_headers.py) (53 lines)

Pure ASGI middleware (not BaseHTTPMiddleware) for maximum performance:

HEADER	VALUE	THREAT MITIGATED
Strict-Transport-Security	max-age=31536000; includeSubDomains; preload	Downgrade attacks, SSL stripping
X-Content-Type-Options	nosniff	MIME sniffing attacks
X-Frame-Options	DENY	Clickjacking
Referrer-Policy	strict-origin-when-cross-origin	Referrer information leakage
Content-Security-Policy	default-src 'self'; img-src 'self' data: https:; style-src 'self' 'unsafe-inline'; script-src 'self'; frame-ancestors 'none'; base-uri 'self'; form-action 'self'	XSS, code injection
X-XSS-Protection	0	Disable legacy XSS auditor (CSP supersedes)

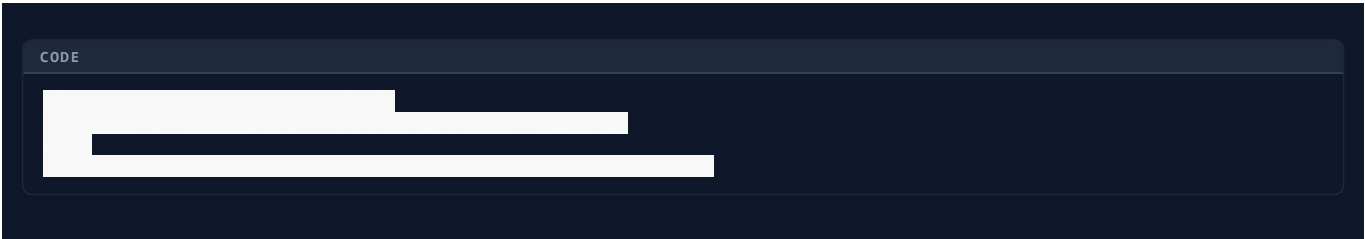
6.5 Cloudflare Proxy Trust

File: [app/core/security_proxy.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/security_proxy.py) (111 lines)

Trusted CIDR Blocks

- 15 Cloudflare IPv4 ranges (e.g., 173.245.48.0/20 , 104.16.0.0/13)
- 7 Cloudflare IPv6 ranges (e.g., 2400:cb00::/32 , 2606:4700::/32)
- Local trusted networks: 127.0.0.0/8 , ::1/128
- Test client bypass: "testclient" and "localhost" hostnames always trusted.

IP Resolution Algorithm



Request Tracing

Every response includes:

- `X-Request-ID` : UUID v4 (from incoming header or auto-generated)
- `CF-Ray` : Cloudflare ray ID (forwarded if present)

6.6 Frontend Security Measures

- **URL Sanitization:** `sanitizeUrl()` in `[ constants.js ]`(`file:///c:/Users/skfir/Desktop/Project-1/src/lib/constants.js`) strips `javascript:` , `data:` , `vbscript:` schemes.
- **HttpOnly Cookie Auth:** Tokens never exposed to JavaScript.
- **Credential Inclusion:** `credentials: 'include'` enforced on every API call.
- **CSP Enforcement:** `script-src 'self'` prevents inline script injection.

Section 7: Testing Architecture

7.1 Test Suite Overview

LAYER	RUNNER	FILES	TEST COUNT
Backend (Python)	pytest + pytest-asyncio	28 test files	143 passed, 1 failed
Frontend Unit	Vitest + Testing Library	16 component tests + 2 unit tests	—
Frontend Integration	Vitest + Testing Library	4 integration tests	—
Frontend E2E	Vitest + Testing Library	2 end-to-end tests	—
Total Frontend	Vitest	24 test files	144 passed

7.2 Backend Test Inventory

TEST FILE	MODULE UNDER TEST	KEY ASSERTIONS
[test_config.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_config.py)	app.core.config	JWT secret entropy, CORS parsing, settings singleton
[test_database.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_database.py)	app.core.database	Engine creation, session lifecycle, SQLite/PG dialect handling
[test_crypto.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_crypto.py)	app.core.security.crypto	bcrypt hashing, JWT creation/decode, token expiration, algorithm whitelist
[test_jwt_service.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_jwt_service.py)	JWT service layer	Access/refresh token creation, payload claims validation
[test_email_validator.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_email_validator.py)	email_filter	RFC validation, disposable domain rejection, normalization
[test_user_model.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_user_model.py)	User model	Column constraints, timezone awareness, relationship cascade
[test_token_model.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_token_model.py)	RefreshToken model	JTI uniqueness, revocation flag, replaced_by chain
[test_auth_models.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_auth_models.py)	Auth model integration	User ↔ RefreshToken cascade behavior
[test_article_schema.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_article_schema.py)	Article model	TSVECTOR generation, GIN index, share_token uniqueness
[test_interaction_models.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_interaction_models.py)	Bookmark / ArticleRead	Unique constraints, cascade deletes, timestamp defaults
[test_article_repository.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_article_repository.py)	ArticleRepository	Batch insert, deduplication, ON CONFLICT DO NOTHING
[test_gnews_client.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_gnews_client.py)	GNewsClient	HTTP mock, rate pacing, 429 handling, timeout handling
[test_auth_register.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_auth_register.py)	Registration endpoints	OTP dispatch, duplicate rejection, password validation
[test_auth_login.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_auth_login.py)	Login endpoints	Credential verification, cookie setting, unverified rejection
[test_auth_guard.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_auth_guard.py)	get_current_user	Token extraction, revocation check, missing token handling

TEST FILE	MODULE UNDER TEST	KEY ASSERTIONS
[test_register_flow.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_register_flow.py)	Full registration flow	Register → OTP verify → Login end-to-end
[test_login_rotation.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_login_rotation.py)	RTR flow	Token rotation, reuse detection, family revocation
[test_routes_feed.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_routes_feed.py)	Feed endpoint	Pagination, category filtering, interaction annotation
[test_routes_search.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_routes_search.py)	Search endpoint	FTS, live fallback, disconnect handling
[test_routes_bookmarks.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_routes_bookmarks.py)	Bookmark endpoints	CRUD, idempotency, ownership enforcement
[test_routes_reads.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_routes_reads.py)	Read status endpoints	Mark/unmark, idempotency, IntegrityError catch
[test_routes_share.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_routes_share.py)	Share endpoint	Token resolution, 404 handling, auth annotation
[test_disconnect_handling.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_disconnect_handling.py)	Search disconnect guards	Client cancellation at 3 guard points
[test_scheduler.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_scheduler.py)	Background poller	Polling cycle, error isolation, graceful shutdown
[test_main_lifespan.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_main_lifespan.py)	Lifespan manager	Startup/shutdown, conditional scheduler
[test_security_headers.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_security_headers.py)	Security headers middleware	All 6 headers present on every response
[test_security_audit.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_security_audit.py)	Penetration tests	IDOR, token tampering, rate limiting, session hijacking
[test_security_idor.py](file:///c:/Users/skfir/Desktop/Project-1/tests/test_security_idor.py)	IDOR attack vectors	Cross-user bookmark access, delete attempts

7.3 Test Infrastructure

Backend [conftest.py](file:///c:/Users/skfir/Desktop/Project-1/tests/conftest.py)

- Registers `to_tsvector()` mock functions on SQLite connections for TSVECTOR compatibility.
- Autouse `reset_login_rate_limiter` fixture clears the global rate limiter before and after every test.

Frontend [setup.js](file:///c:/Users/skfir/Desktop/Project-1/tests/setup.js)

JAVASCRIPT

```
import '@testing-library/jest-dom'
```

Vitest Configuration (in [vite.config.js](file:///c:/Users/skfir/Desktop/Project-1/vite.config.js))

JAVASCRIPT

```

: true
: 'jsdom'
: './tests/setup.js'
: 'tests/**/*.test.{js,jsx}'

```

pytest Configuration (in [pyproject.toml](file:///c:/Users/skfir/Desktop/Project-1/pyproject.toml))

TOML

```
[tool.pytest.ini_options]
    = "tests"
    = "."
    = "auto"
```

7.4 Frontend Test Inventory

CATEGORY	TEST FILES	COVERAGE
Component Tests (16)	ArticleCard , ArticleGrid , AuthPosterLayout , BookmarksDrawer , CategoryFilter , HeroArticle , LoginForm , Masthead , OtpVerification , RegisterForm , SearchBar , SwissBadge , SwissButton , SwissErrorBoundary , SwissGrid , SwissInput	Rendering, user interaction, props, variants
Page Tests (1)	SharePage	Token resolution, error states, loading states
Unit Tests (2)	api.test.js , constants.test.js	Fetch wrapper, URL sanitization, route constants
Integration Tests (4)	api_client.test.js , network_sync.test.jsx , use_article_interactions.test.js , use_news_feed.test.js	Hook behavior, optimistic updates, retry logic, abort handling
E2E Tests (2)	integration_resilience.test.jsx , security_and_flow.test.jsx	End-to-end flows, resilience scenarios, security assertions

Section 8: External Services & Integrations

8.1 GNews API Client

File: [app/services/gnews_client.py](file:///c:/Users/skfir/Desktop/Project-1/app/services/gnews_client.py) (137 lines)

METHOD	ENDPOINT	PARAMETERS	BEHAVIOR
fetch_top_headlines(category)	GET /top-headlines	category, country=in, lang=en, max=10, apikey	Sequential with 1.2s inter-request delay
fetch_categories(categories)	—	List of category strings	Sequential multi-category fetch
search_articles(query)	GET /search	q, in=title, country=in, lang=en, max=10, apikey	Full-text upstream search

Rate Pacing: `asyncio.Lock` -protected throttle enforcing 1.2-second minimum delay between consecutive requests.

Error Handling:

- HTTP 429 → graceful empty return (no exception raised).
- `httpx.TimeoutException` / `httpx.RequestError` → warning log, empty return.
- All exceptions caught; never propagated to caller.

8.2 Article Repository

File: [app/services/article_repository.py](file:///c:/Users/skfir/Desktop/Project-1/app/services/article_repository.py) (180 lines)

`normalize_article_data()` — Raw JSON → Schema-Compliant Dict

- Resolves `published_at` from `publishedAt` (GNews format) or `published_at`, with ISO 8601 parsing and `Z` → `+00:00` replacement.
- Resolves `source_name` from nested `source.name` (GNews format) or flat `source_name`.
- Generates `share_token` via `secrets.token_urlsafe(8)` → 11 base64 characters, truncated to 16 chars.
- Truncates fields: `title[:500]`, `url[:1000]`, `source_name[:100]`, `category[:50]`, `country[:10]`, `language[:10]`, `share_token[:16]`.

Batch Insert Strategy

CODE

- **PostgreSQL:** `sqlalchemy.dialects.postgresql.insert(...).on_conflict_do_nothing(index_elements=["url"])`
- **SQLite:** `sqlalchemy.dialects.sqlite.insert(...).on_conflict_do_nothing(index_elements=["url"])`
- **Intra-batch deduplication:** `_deduplicate_batch()` filters duplicate URLs within the same batch to avoid multi-row intra-statement conflicts.

Dual Session Support

METHOD	SESSION TYPE	USE CASE
batch_insert_async()	AsyncSession	Background scheduler
batch_insert_sync()	Session	Search live-fallback (sync within async route)
batch_insert()	Union dispatch	Unified entry point

8.3 Email Service

File: [app/services/email_service.py](file:///c:/Users/skfir/Desktop/Project-1/app/services/email_service.py) (50 lines)

Dual-mode email dispatch:

1. **SMTP Mode:** When `SMTP_HOST` is not `localhost` / `127.0.0.1` /empty or `SMTP_USER` is configured, sends via `smtplib.SMTP` with STARTTLS and optional authentication.
2. **Local Mode:** Records OTP to in-memory `sent_verification_emails` list (accessible by test harness for verification).

All emails are always logged to the in-memory dispatch audit queue regardless of SMTP mode.

Section 9: Configuration, Environment & DevOps

9.1 Application Configuration

File: [app/core/config.py](file:///c:/Users/skfir/Desktop/Project-1/app/core/config.py) (98 lines)

Settings Model (Pydantic v2 BaseSettings)

VARIABLE	TYPE	DEFAULT	PURPOSE
DATABASE_URL	str	postgresql+asyncpg://postgres:postgres@localhost:5432/signalreport	Database connection string
DB_POOL_SIZE	int	20	Connection pool size
DB_MAX_OVERFLOW	int	10	Maximum overflow connections
DB_POOL_TIMEOUT	int	30	Pool acquisition timeout (seconds)
GNEWS_API_KEY	str	""	GNews API key
GNEWS_BASE_URL	str	https://gnews.io/api/v4	GNews API base URL
JWT_SECRET_KEY	str	required	HMAC signing key (≥256 bits enforced)
JWT_ALGORITHM	str	HS256	JWT signing algorithm
ACCESS_TOKEN_EXPIRE_MINUTES	int	15	Access token TTL
REFRESH_TOKEN_EXPIRE_DAYS	int	14	Refresh token TTL
CORS_ORIGINS	`list[str]`	str`	["http://localhost:3000", "http://localhost:8000"]
COOKIE_SECURE	bool	False	Cookie Secure flag
COOKIE_SAMESITE	str	lax	Cookie SameSite policy
POLLING_INTERVAL_HOURS	int	2	Background polling interval
SMTP_HOST	str	localhost	SMTP mail server
SMTP_PORT	int	587	SMTP port
EMAIL_FROM	str	noreply@signalreport.io	Sender email address

Validators

- `JWT_SECRET_KEY` : Enforces `len(value.encode("utf-8")) >= 32` (256 bits minimum entropy for HMAC-SHA256).
- `CORS_ORIGINS` : Parses comma-separated strings (`"http://a,http://b"`) and JSON arrays (`'["http://a","http://b"]'`).

Singleton Pattern

`get_settings()` caches a single `Settings` instance in module-level `_settings` variable.

9.2 Environment File

File: [.env](file:///c:/Users/skfir/Desktop/Project-1/.env) (git-ignored)

Contains: `DATABASE_URL` , `JWT_SECRET_KEY` , `GNEWS_API_KEY` , `CORS_ORIGINS` , `ENABLE_SCHEDULER` , and SMTP configuration.

9.3 Build & Development Tooling

Backend

TOOL	COMMAND	PURPOSE
Uvicorn	<code>uvicorn app.main:app --reload --port 8000</code>	Development server
pytest	<code>pytest tests/</code>	Run backend test suite
pip	<code>pip install -r requirements.txt</code>	Install Python dependencies

Frontend

TOOL	COMMAND	PURPOSE
Vite Dev Server	<code>npm run dev</code>	Development server (port 5173)
Vite Build	<code>npm run build</code>	Production bundle
Vitest	<code>npm run test</code>	Run frontend test suite

Proxy Configuration

[vite.config.js](file:///c:/Users/skfir/Desktop/Project-1/vite.config.js) proxies `/api` requests to `http://127.0.0.1:8000` during development.

9.4 Python Dependencies

File: [requirements.txt](file:///c:/Users/skfir/Desktop/Project-1/requirements.txt) (16 entries)

PACKAGE	VERSION CONSTRAINT	PURPOSE
<code>fastapi</code>	<code>≥0.111.0</code>	ASGI framework
<code>uvicorn</code>	<code>≥0.29.0</code>	ASGI server
<code>sqlalchemy</code>	<code>≥2.0.0</code>	ORM
<code>asyncpg</code>	<code>≥0.29.0</code>	Async PostgreSQL driver
<code>aiosqlite</code>	<code>≥0.20.0</code>	Async SQLite driver (tests)
<code>httpx</code>	<code>≥0.27.0</code>	Async HTTP client
<code>pydantic</code>	<code>≥2.7.0</code>	Data validation
<code>pydantic-settings</code>	<code>≥2.2.0</code>	Settings management
<code>pyjwt</code>	<code>≥2.8.0</code>	JWT handling
<code>cryptography</code>	<code>≥42.0.0</code>	Cryptographic primitives
<code>bcrypt</code>	<code>≥4.0.1</code>	Password hashing
<code>passlib[bcrypt]</code>	<code>≥1.7.4</code>	Password utility (legacy compat)
<code>email-validator</code>	<code>≥2.1.0</code>	RFC email validation
<code>pytest</code>	<code>≥8.0.0</code>	Test runner
<code>pytest-asyncio</code>	<code>≥0.23.0</code>	Async test support

9.5 Node.js Dependencies

File: [package.json](file:///c:/Users/skfir/Desktop/Project-1/package.json)

Runtime (2): `react@^18.3.1`, `react-dom@^18.3.1`
Dev (8): `@vitejs/plugin-react`, `vite`, `vitest`, `@testing-library/react`, `@testing-library/jest-dom`, `@testing-library/user-event`, `tailwindcss`, `autoprefixer`, `postcss`, `jsdom`

9.6 Complete File Manifest

Backend Python Files

FILE	PATH	LINES	BYTES
<code>main.py</code>	<code>app/main.py</code>	95	3,332
<code>config.py</code>	<code>app/core/config.py</code>	98	3,284
<code>database.py</code>	<code>app/core/database.py</code>	95	2,729
<code>security_headers.py</code>	<code>app/core/security_headers.py</code>	53	1,922
<code>security_proxy.py</code>	<code>app/core/security_proxy.py</code>	111	3,803
<code>scheduler.py</code>	<code>app/core/scheduler.py</code>	111	4,209
<code>crypto.py</code>	<code>app/core/security/crypto.py</code>	152	4,384
<code>email_filter.py</code>	<code>app/core/security/email_filter.py</code>	120	3,734
<code>rate_limiter.py</code>	<code>app/core/security/rate_limiter.py</code>	101	3,682
<code>deps.py</code>	<code>app/api/deps.py</code>	106	3,512
<code>auth_register.py</code>	<code>app/api/v1/auth_register.py</code>	132	4,211
<code>auth_login.py</code>	<code>app/api/v1/auth_login.py</code>	280	8,776
<code>routes_feed.py</code>	<code>app/api/v1/routes_feed.py</code>	133	3,995
<code>routes_search.py</code>	<code>app/api/v1/routes_search.py</code>	229	7,964
<code>routes_bookmarks.py</code>	<code>app/api/v1/routes_bookmarks.py</code>	322	9,270
<code>routes_reads.py</code>	<code>app/api/v1/routes_reads.py</code>	219	6,042
<code>routes_share.py</code>	<code>app/api/v1/routes_share.py</code>	108	3,376
<code>user.py</code>	<code>app/models/user.py</code>	34	1,308
<code>token.py</code>	<code>app/models/token.py</code>	30	1,114
<code>article.py</code>	<code>app/models/article.py</code>	91	3,180
<code>interactions.py</code>	<code>app/models/interactions.py</code>	80	2,242
<code>gnews_client.py</code>	<code>app/services/gnews_client.py</code>	137	5,296
<code>article_repository.py</code>	<code>app/services/article_repository.py</code>	180	6,066
<code>email_service.py</code>	<code>app/services/email_service.py</code>	50	1,981

Frontend Source Files

FILE	PATH	BYTES
<code>App.jsx</code>	<code>src/App.jsx</code>	4,980
<code>main.jsx</code>	<code>src/main.jsx</code>	252
<code>api.js</code>	<code>src/lib/api.js</code>	5,988
<code>constants.js</code>	<code>src/lib/constants.js</code>	1,408

FILE	PATH	BYTES
useNewsFeed.js	src/hooks/useNewsFeed.js	5,509
useArticleInteractions.js	src/hooks/useArticleInteractions.js	5,140
useNetworkSync.js	src/hooks/useNetworkSync.js	3,268
DashboardPage.jsx	src/pages/DashboardPage.jsx	10,117
SharePage.jsx	src/pages/SharePage.jsx	7,678
globals.css	src/styles/globals.css	725

Section 10: Known Technical Debt & Remediation Roadmap

10.1 Critical Issues

10.1.1 Sync/Async Session Type Mismatch

CAUTION

****Severity**:** Critical — affects all route handlers in production.

****Problem**:** `[get_db()]`(file:///c:/Users/skfir/Desktop/Project-1/app/core/database.py#L85-L94) yields ``AsyncSession``, but all route handlers and `[deps.py]`(file:///c:/Users/skfir/Desktop/Project-1/app/api/deps.py#L4) type-hint ``Session`` (synchronous) and use ``db.query()`` — the legacy SQLAlchemy 1.x API that does not exist on ``AsyncSession``.

Impact: Calling `db.query(User).filter_by(...)` on an `AsyncSession` raises `AttributeError: 'AsyncSession' object has no attribute 'query'` at runtime.

Root Cause: The migration from synchronous SQLAlchemy to async was incomplete. Route handlers were not refactored to use the SQLAlchemy 2.0 async query API (`await session.execute(select(...))`).

Remediation:

DIFF

```
- from sqlalchemy.orm import Session
+ from sqlalchemy.ext.asyncio import AsyncSession

- def get_current_user(request, db: Session = Depends(get_db)):
+ async def get_current_user(request, db: AsyncSession = Depends(get_db)):
-     user = db.query(User).filter_by(id=user_id).first()
+     result = await db.execute(select(User).filter_by(id=user_id))
+     user = result.scalar_one_or_none()
```

This refactor must be applied across **all 7 route files** and `deps.py`.

10.1.2 CORS Middleware Not Configured

Problem: `[app/main.py]`(file:///c:/Users/skfir/Desktop/Project-1/app/main.py) does not add `CORSMiddleware` from Starlette, despite `CORS_ORIGINS` being configured in `Settings`. This means cross-origin browser requests from the Vite dev server (port 5173) to FastAPI (port 8000) will be blocked by browser CORS policies.

Current Workaround: Vite proxy (`/api → http://127.0.0.1:8000`) bypasses CORS in development, but production deployments without a proxy will fail.

Remediation: Add `CORSMiddleware` to `main.py`:

PYTHON

```
from starlette.middleware.cors import CORSMiddleware

...

= True
= """
= """
```

10.2 High-Priority Issues

10.2.1 Test Config Assertion Failure

`[test_config.py]`(file:///c:/Users/skfir/Desktop/Project-1/tests/test_config.py) contains a hardcoded assertion that `CORS_ORIGINS` equals `["http://localhost:3000", "http://localhost:8000"]`, but the `.env` file was updated to include `http://localhost:5173` for Vite compatibility,

10.5 Remediation Priority Matrix

PRIORITY	ITEM	EFFORT	IMPACT
P0	Async session refactor (all routes + deps)	2-3 days	Blocks all production API calls
P0	Add CORSMiddleware	15 minutes	Blocks cross-origin production deployment
P1	Fix test_config.py assertion	10 minutes	Unblocks CI pipeline
P1	Add created_at to RefreshToken model	20 minutes	Enables RTR timestamp validation
P2	Integrate Alembic migrations	1 day	Production schema management
P2	Remove passlib dependency	5 minutes	Dependency hygiene
P3	Clean .gitignore	10 minutes	Repository hygiene
P3	Consolidate dual router mounts	30 minutes	API surface clarity
P3	Migrate to cursor pagination	1-2 days	Deep-page performance

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